Friction-Stir Welding Machine

Concordia University Engineering and Computer Science

Presented by: Team 2

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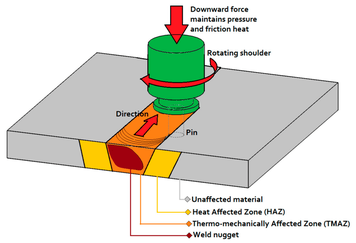
Background

Friction-Stir Welding (FSW) is a form of Solid-State Welding that applies a circular friction and axial force to effectively "Mix" the two interacting surfaces with heat. This is achieved by using a specialized tool that is designed to generate heat and penetrate the joining surfaces to create a weld "Pool".

This project focuses on developing and fabricating a FSW machine with a compact design that features a 40-inch cubic volume and is appropriate for a desktop. The FSW machine combines multiple parameters to perform the welding process successfully. The challenge is combining all the parameters from the literature review to modify a mini-computer numerical controlled (CNC) machine into an FSW machine.

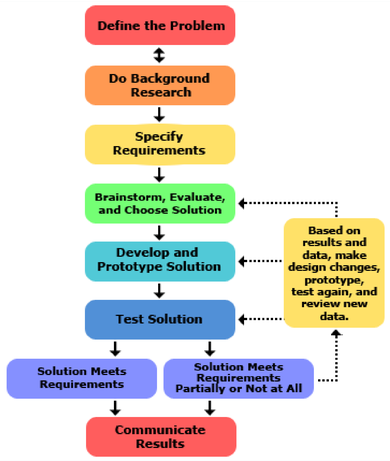


Project Scope

Objectives:

1. Design and Manufacture a Friction-Stir Welding Machine
2. Weld Two Similar Metals Together

Process Plan



Specifications

Material: 6061-T5 Aluminum Alloy

Type of Weld: Butt Weld

Welding Speed: 1500 RPM

Plunge Rate: 0.1 in/min

Plunge Depth: 0.1 inch

Feed Rate: 2 in/min

Axial Force: 4.7 kN

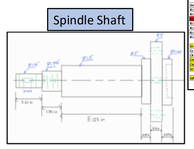
Transverse Force: 2 kN

Minimum Required Torque: 5 N.m.

Minimum Required Motor Power: 1.02 HP

Preliminary Calculations

Spindle Shaft



Material	6061-T6 (-T651/-T652)
Tensile Strength (Sut)	310 MPa
Yield Strength (Sy)	276 MPa
Fatigue Strength (Se)	96.5 MPa
Stress Concentration Kt (Bending)	1.5
Kts (Torsion)	1.0
Fatigue Stress Concentration Kf (Bending)	1.4
Kfs (Torsion)	1.0
Kt (Axial)	1.5
Kt (Shear)	1.0
Kt (Profile key)	2.0
Ks (Surface - Ground)	0.89
Kb (Size factor)	1.0
Kc (Reliability factor)	0.814 (99.99% reliability)
Kd (Temperature factor)	1.0
Ke (Miscellaneous effects factor)	1.0
Safety factor	2.0

Forces

Common Force throughout the shaft

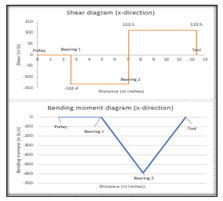
Axial (z)	4000 N	
	899.2358 lb	
Welding (y)	2000 N	
	449.6179 lb	
Transverse (x)	500 N	
	112.4045 lb	
Compression due to axial force	1144.941 psi	
	7.896155 Mpa	

Bearing 1 (below pulley)

Kf (Fluctuation factor)	1.0
C (Supplied shaft load)	899.22077 lb
S (Applied thrust load)	0.56
K (Thrust factor)	1.5
L (Desired hours of run time)	5000
n	1500
Kn	3 (for ball bearings)
Design Life (L d)	39000000 cycles
Fl (equivalent load)	934.6703 lb
C (Basic dynamic load ra	**5373.97 lb**

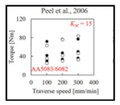
Bearing 2 (below bearing 1)

Kf (Fluctuation factor)	1.0
C (Supplied shaft load)	361.02832 lb
S (Applied thrust load)	899.22077 lb
K (Thrust factor)	0.56
L (Desired hours of run time)	5000
n	1500
Kn	3 (for ball bearings)
Design Life (L d)	3.90E+07 cycles
Fl (equivalent load)	934.2195 lb
C (Basic dynamic load ra	**6403.32 lb**



Stirring Tool

Tool Geometry		inch
Ds (mm)	6.35	1/4"
Dx (mm)	12.7	1/2"
Pt (mm)	2.54	0.1"
G (mm^2)	114.0092	



Torque	
Km	15
G	114
ω	1500
v	50
t	7
M (N-m)	5

$$G = \frac{\pi}{4} D_p^2 + \pi D_p p_1 + \frac{\pi}{4}(D_s - D_p)^2$$

$$M = K_M \frac{G}{\omega} \sqrt[3]{vt}.$$

Temperature Constant	
G	114
ω	1500
v	50
t	7
Ct	9486

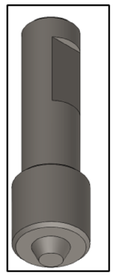
$$C_M = \frac{G}{\omega} \sqrt[3]{vt}.$$

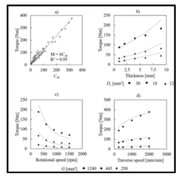
and,

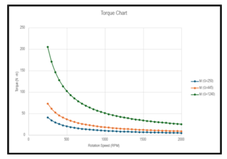
$$C_T = \frac{G\omega}{\sqrt[3]{vt}}$$

Max Temperature	
Kt	50
Ct	9486
PHI	0.25
T	493

$$\begin{cases} T = K_T C_T^2 & \text{for } C_T < 20000, \\ T = 590^\circ C & \text{for } C_T \geq 20000. \end{cases}$$



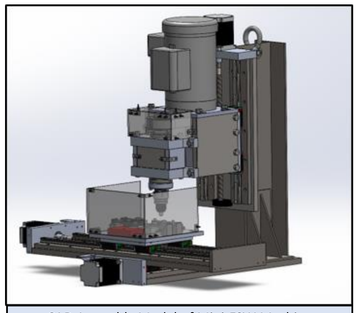




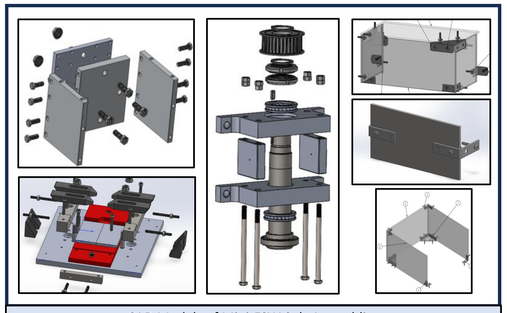
Pulley Inp (DIA)	65.03
Pulley Out (DIA)	74.93
Motor Sample	
HP&RPM to T	
HP	1.5
RPM	1800
Torque	5.93
Gear Reduction	
Ratio	1.15
Output Torque	6.84
Output RPM	1562.18

Motor Specs		
Torque (Nm)	4.840727917	
RPM	1500	HP
kW	0.76040338	1.019701

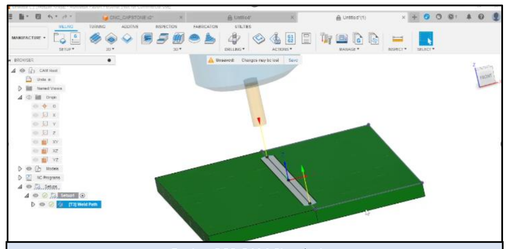
Design



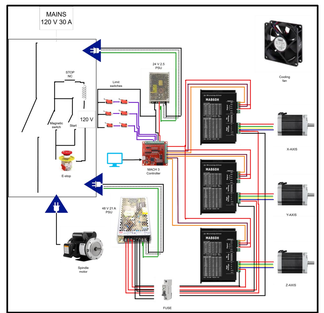
CAD Assembly Model of Mini-FSW Machine



CAD Models of Mini-FSW Sub-Assemblies

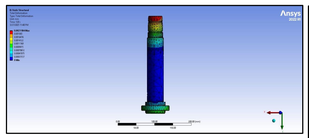


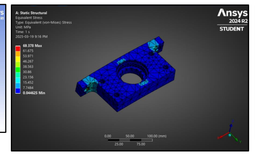
Fusion 360 CAM Simulation

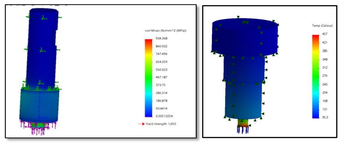


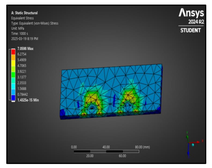
Wiring layout for CNC Integration

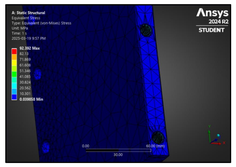
FEA and Testing

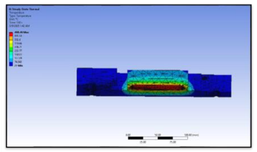


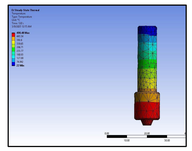






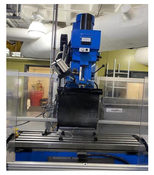




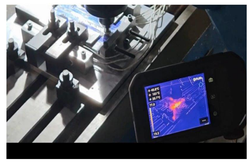


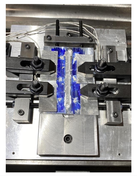
Testing on 6061-T5 Aluminum Alloy Sample:

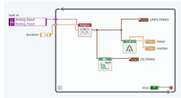
1) Collection of temperature readings using a thermal camera and 6 K-type thermocouples



Fagor 8050 machine used for weld test







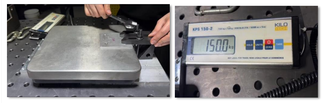
DAQExpress block diagram for data acquisition

2) Grain structure at the weld site

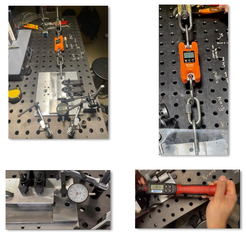
3) Tensile test sample - welded sample (left) and stock sample (right)

Tests on the Clamping System:

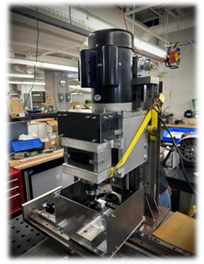
1) Clamp Force Test using Weight Scale



2) Clamping System Test using Force Gauge



Results

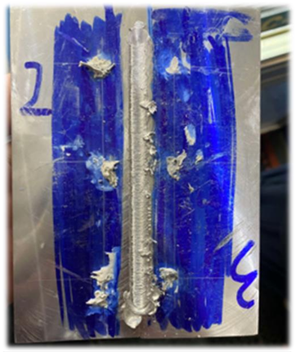


Mini-FSW Machine

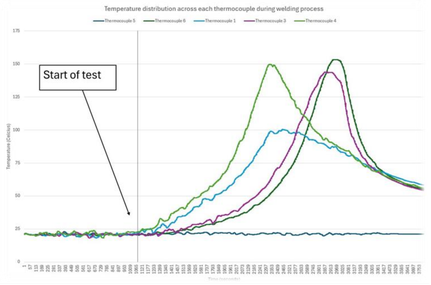
Stirring Tools

Stirring Tool After Test

Spindle Shaft



FSW Sample Plate



Temperature Readings During FSW Test

Conclusions and Future Directions

Conclusions:

1. The welding operation, clamping system and the stirring tool were designed to withstand the applied stresses with no visible defects.
2. The installation method of the thermocouples in the workpiece must be reviewed, as the probe ends of the thermocouples vary, preventing them from being properly positioned in the plate. This resulted in the temperature readings being more erroneous.

Future Directions:

1. Perform welding test using different pin geometries and welding parameters to weld 6061-T5 aluminum sample (0.25" in thickness) with the least number of defects.
2. Determining the welding parameters to weld dissimilar metals.
3. Perform a more in-depth grain analysis by using the right chemical for etching the cross section of the weld
4. Perform tensile tests on the welded and stock samples and compare them to the theoretical properties of 6061 aluminum to validate the integrity of the weld.

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- CDC Members
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- Faculty of Dawson College